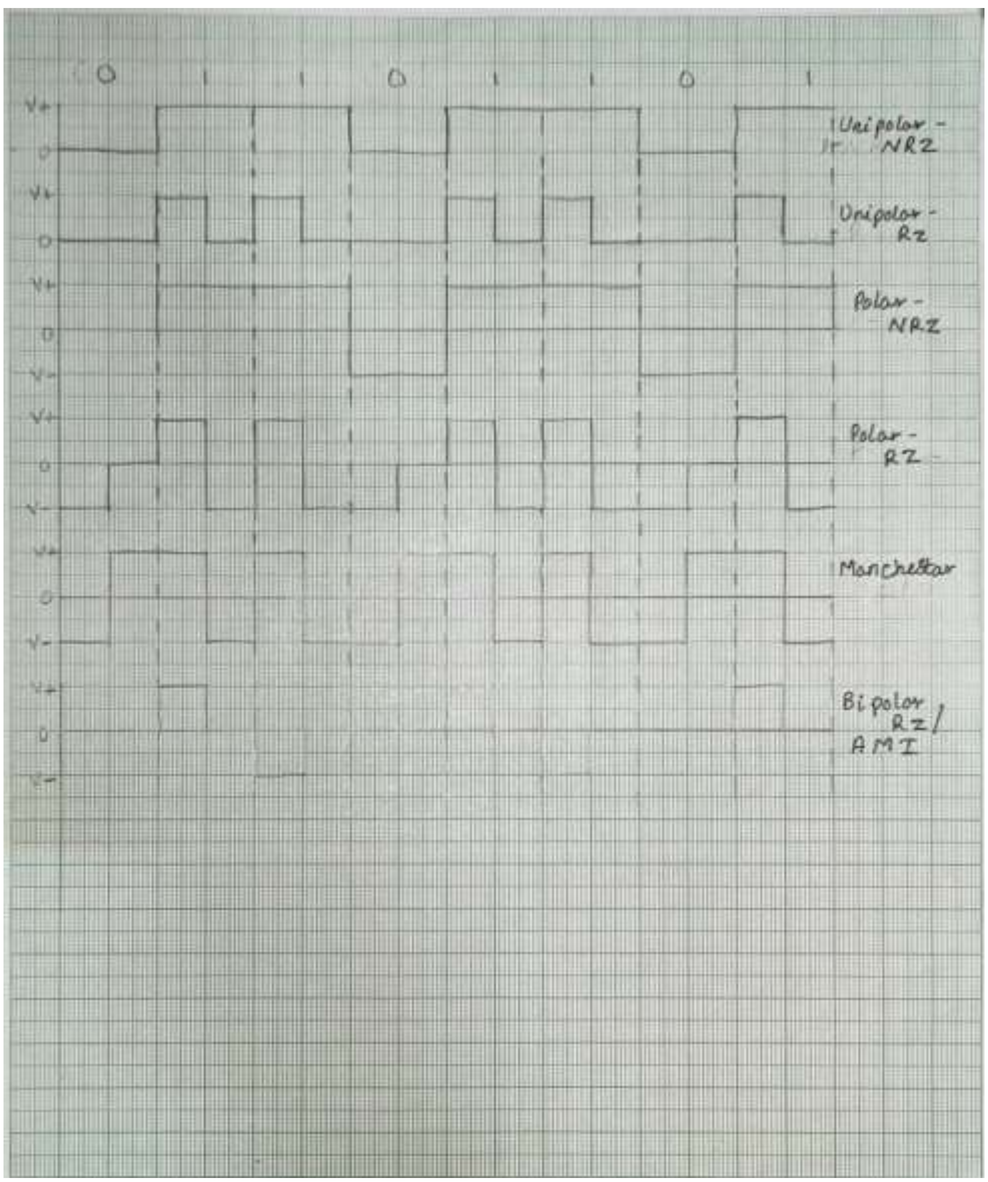
Mahavir Education Trust's
Shah & Anchor Kutchhi Engineering College
An Autonomous Institute Affiliated to University of Mumbai

UG Program in Electronics and Communication
(Advanced Communication Technology)

Experiment No. 1
Generation of Line Codes

1.1 Aim: To study complete circuit of Line Coding and Decoding and observe waveforms on DSO

1.2 Line Coding waveforms:



1.3 Conclusion:

In this experiment, different line coding techniques such as Unipolar NRZ, Unipolar RZ, Polar NRZ, Polar RZ, Manchester, and Bipolar (AMI) were studied. The generation and decoding circuits of line codes were analyzed and the corresponding waveforms were observed on the DSO. It was observed that each line code has different characteristics in terms of DC component, bandwidth, synchronization, and error detection capability. Thus, line coding plays an important role in reliable digital data transmission.

1.4 Questions:

1. State the Important properties of Line Codes

The important properties of line codes are:

- **DC Component:** Line code should have minimum or zero DC component.
- **Bandwidth Efficiency:** It should require less transmission bandwidth.
- **Synchronization:** The receiver should be able to extract clock information easily.
- **Power Efficiency:** Average transmitted power should be low.
- **Error Detection Capability:** Some line codes help in detecting errors.
- **Noise Immunity**: It should be less affected by noise and distortion.
- **Simplicity:** Encoding and decoding circuits should be simple.

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2.Compare Different Line Codes.

Line Code	Signal Levels	DC Balance	Bandwidth	Synchronization	Error Detection
Unipolar NRZ	0 or +V	Poor (DC bias)	Low (bit rate)	Poor (long runs possible)	None
Polar NRZ	+V or -V	Good	Low (bit rate)	Poor	Low
Unipolar RZ	0 (full) or +V (half)	Poor	Higher (2x bit rate)	Better	Low
Bipolar RZ (AMI)	+V/-V (alternate) or 0	Excellent	Higher (2x bit rate)	Good	High (violations detectable)
Manchester	+V to -V or -V to +V mid-bit	Excellent	High (2x bit rate)	Excellent (mid-bit transition)	Medium